

the next **PANGEA**

**APPLIED
ARTIFICIAL
INTELLIGENCE
AND DIGITAL
DEVELOPMENTS**

IMAGIA

IMAGE ANALYSIS

Digital platform based on Artificial Intelligence for image analysis and defect detection

The application of Artificial Intelligence based on image processing is becoming increasingly common to automate various everyday tasks. TNP has a specialized section dedicated to developing solutions based on Deep Learning for the resolution and optimization of complex industrial cases that are not easily addressed by conventional models.

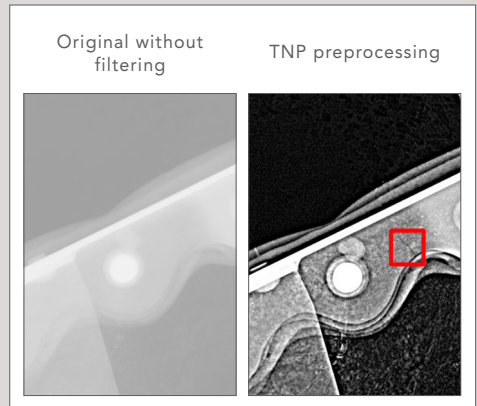
One of the primary products is IMAGIA, a **platform that accelerates labeling and automation of customized models** using state-of-the-art algorithms based on supervised and unsupervised techniques, supported by weak or noisy labels. The key advantages of these developments include:

Precision & Automation

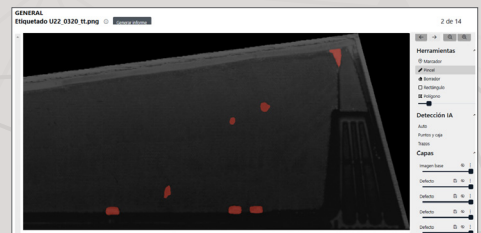
These algorithms achieve **high accuracy in detecting objects, patterns, and anomalies** in complex industrial data by analyzing a large volume of data.

Specialization

These models adapt to the data to meet the specific needs of different industrial



Sample of automatic preprocessing of an X-ray



Component with defects automatically detected through unsupervised methods using TNP Software

environments, allowing the development of **highly specialized solutions to address industrial challenges.**

AI Lifecycle

This type of solution enables continuous improvement as more data is collected and models are refined, leading to a **constant increase in the accuracy and effectiveness of the solutions.**

Decision Support

Having data-driven information supports strategic decision-making, aiding in taking proactive measures to enhance quality and efficiency.

PLANIFY

SUPPLY CHAIN OPTIMIZATION

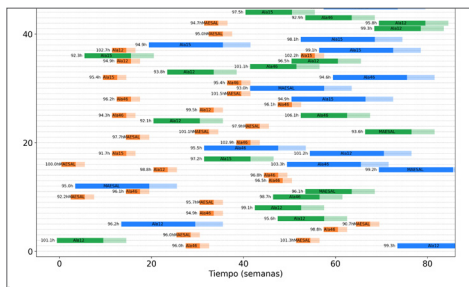
Advanced and Intelligent Supply Chain Optimization: Strategy, Prediction, and Planning

This product represents a significant transformation in supply chain management, integrating traditional techniques with the power of the most advanced artificial intelligence algorithms. These algorithms are deployed in three critical areas: Prediction, to anticipate variations and trends with greater accuracy than current methods; Strategy, where large volumes of data are processed and analyzed to chart effective courses of action; and Planning, optimizing both high and low-level processes, ensuring that every step in the chain is as efficient as possible.

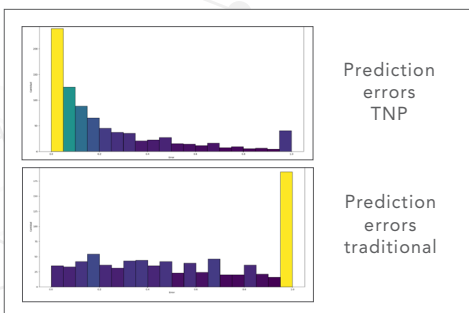
The adoption of this technological triad not only perfects each link in the supply chain but also translates into tangible benefits for businesses: from **agility in decision-making** to a **significant reduction in operational expenses**. In summary, this product offers a comprehensive solution that positions organizations several steps ahead in an increasingly competitive and changing market environment.

Accuracy and anticipation

The ability to anticipate variations and trends with a higher accuracy than conventional methods, achieving high



Capture of the management platform, displaying maintenance insights for each aircraft (each aircraft represented as a horizontal strip with corresponding indicators)



Measured prediction errors of consumption for over 1500 items with different demand patterns

efficiency in measuring the error between predictions and actual values. **Analyzing large volumes of data enables the formulation of more effective strategies.**

Process optimization

Average reduction in the time required to complete specific processes, thanks to the refinement and improvement of both high and low-level processes. Improved adaptability to changes, with a substantial enhancement in response times compared to traditional methods.

Planning and costs

Improvement in planning and competitiveness, enabling cost reduction through optimized decision-making.

DEXTER

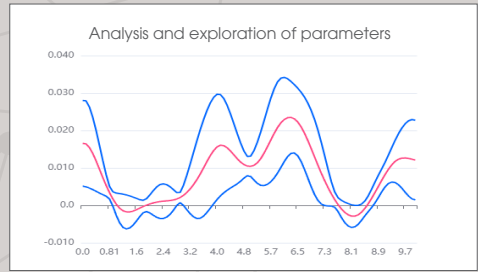
PARAMETRIC OPTIMIZATION

Intelligent parameter exploration and adjustment in experimental research environments

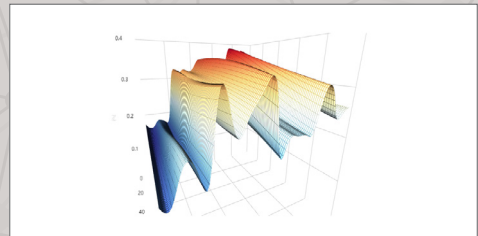
This product stands at the forefront of experiment optimization and management in research environments. It blends traditional approaches with the precision of cutting-edge artificial intelligence-based tools. The product focuses on two fundamental pillars: Exploration, which surveys a broad range of parameters and variables to discover highly relevant opportunities, and Optimization, where advanced techniques are deployed to ensure experiments yield the most favorable results with efficient resource utilization.

Experiment Exploration and Optimization enable simulations of real processes with great accuracy. **These precise simulations allow researchers to reduce the number of experiments needed to reach the optimal solution.**

By integrating this product into their processes, research environments will not only enhance the accuracy and reliability of their experiments but also benefit from more agile management and a significant reduction in operational costs.



Graphic representation of the confidence value in the solution, where the solution value is on the Y-axis and the input variable values are on the X-axis (confidence intervals in blue)



Visualization of the reduced three-dimensional solution search space (in blue, values closer to the optimum)

Efficient optimization

Significant reduction in the average number of experiments needed to reach an optimal solution, thanks to the refinement and adjustment of experimental parameters. **Optimization enables a reduction in operational costs** by decreasing the number of experiments and calculations compared to traditional systems.

The software allows a comprehensive exploration of variables and parameters to **discover combinations with high potential while accelerating the parameter selection process** and experiment design based on anticipated optimal results.

